

# Option Pricing: Motivation and First Mathematical Tools

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# How the course materials fit together

## Slides

The in-class : core ideas, examples, and formulas.

## Lecture notes

The full reference: definitions, derivations, proofs, and worked examples.

## Exercises and HW

The practice layer: calculations, code...

# Homework and final grade

## Homework rhythm

- Homework is assigned **weekly**; you have **two weeks** to complete each assignment. Brightspace gives the official deadline and submission instructions.
- **Practice assignments:** mathematical reasoning and calculations.
- **Numerical assignments:** simulation or approximation, with code and interpretation.
- The **lowest homework score is dropped**.

## Letter-grade scale

| F       | D     | D+    | C-    | C     | C+    | B-    | B     | B+    | A-    | A     |
|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| < 62.45 | 62.45 | 66.45 | 69.45 | 72.45 | 76.45 | 79.45 | 82.45 | 86.45 | 89.45 | 94.45 |

*Each value indicates the minimum percentage required for that letter grade.*

## Two grading routes

|                  | Grade 1 | Grade 2 |
|------------------|---------|---------|
| Homework average | 50%     | 60%     |
| Midterm exam     | 20%     | 0%      |
| Final exam       | 30%     | 40%     |

Final Grade =  $\max(\text{Grade 1}, \text{Grade 2})$

The more favorable calculation is used automatically.

# References (see Brightspaces)

## Main references

- S. Shreve, *Stochastic Calculus for Finance II*, Chapters 1–2.
- I. Karatzas and S. Shreve, *Brownian Motion and Stochastic Calculus*, Chapter 1.
- S. Shreve, *Stochastic Calculus for Finance I*, Chapters 1–2 for discrete-time pricing intuition.

Motivation, replication, and pricing

Probability toolbox

Information and conditional expectation

Processes and martingales

Short differential-calculus review

# Motivation, replication, and pricing

Motivation, replication, and pricing

Contracts and time value

The casino reveals the role of replication

A one-period financial market

- An airline is exposed to future fuel prices; an exporter is exposed to exchange rates.

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## Derivatives reshape future financial risk

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### The pricing question

What amount paid **today** is consistent with a possibly random **future payoff**?

## One option introduces most of the vocabulary

Suppose a one-year call option on a stock has strike  $K = 100$ :

$$X = (S_T - K)^+ = (S_T - 100)^+.$$

- $S$  is the **underlying**;  $T$  is the **maturity**.

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- The **long** pays the option price today and receives  $X$  at  $T$ .

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- $X$  is the **payoff**: its rule is known today and its value is realized at  $T$ .
- The **long** pays the option price today and receives  $X$  at  $T$ .
- The **short** receives the price today and owes  $X$  at  $T$ .

If  $S_T = 125$ , then  $X = 25$ ; if  $S_T = 90$ , then  $X = 0$ .

# A dollar today and a dollar tomorrow are different

At a constant continuously compounded rate  $r$ , the bank account is  $B_t = e^{rt}$ .

## Discounting

A certain payment of  $C$  at time  $T$  has value

$$Ce^{-rT}$$

today, because this amount invested in the bank account produces exactly  $C$  at  $T$ .

## Example

If  $C = 100$ ,  $r = 5\%$ , and  $T = 1$ , the present value is

$$100e^{-0.05} \approx 95.12.$$

## Historical expectation is a natural first guess

I sell a contract with payoff

$$X = \begin{cases} \$100, & \text{if red occurs,} \\ \$0, & \text{if black occurs.} \end{cases}$$

Under the historical probability measure  $\mathbb{P}$ ,

$$\mathbb{P}(\text{red}) = 0.99, \quad \mathbb{P}(\text{black}) = 0.01,$$

so

$$\mathbb{E}^{\mathbb{P}}[X] = 0.99 \times \$100 + 0.01 \times \$0 = \$99.$$

### Pricing question

Is \$99 the arbitrage-free price? **No.** Historical probabilities describe likelihoods; they do not by themselves price a replicable claim.



## A \$50 bet reproduces the contract state by state

What payoff does \$50 placed on red produce?

|                 | Red                     | Black |
|-----------------|-------------------------|-------|
| Contract        | \$100                   | \$0   |
| \$50 bet on red | $2 \times \$50 = \$100$ | \$0   |

same terminal payoff in every state  $\implies$  replication cost = \$50.

### Takeaway

**PRICE = COST OF REPLICATION**

# No arbitrage forces the price to be \$50

$$C_0 = \$50, \quad \text{not} \quad \mathbb{E}^{\mathbb{P}}[X] = \$99.$$

**If the contract costs more than \$50**

- sell the contract;
- buy the \$50 hedge;
- keep the difference with no terminal risk.

**If it costs less than \$50**

- buy the contract;
- sell the replicating bet;
- keep the reverse difference.

## Transition

Historical probabilities describe likelihoods. Which probabilities represent prices consistently with traded assets?

## A one-period stock model makes replication visible

Assume a zero interest rate. A stock is worth  $S_0 = 50$  today and

$$S_1 = \begin{cases} 60 & \text{up,} \\ 40 & \text{down.} \end{cases}$$

A call with strike  $K = 50$  pays

$$X = (S_1 - 50)^+ = \begin{cases} 10 & \text{up,} \\ 0 & \text{down.} \end{cases}$$

Can stock and cash reproduce both payoffs?

## Worked example: replication determines the call price

Hold  $\Delta$  shares and  $b$  dollars in cash. Matching both terminal payoffs gives

$$60\Delta + b = 10, \quad 40\Delta + b = 0.$$

Therefore

$$\Delta = \frac{1}{2}, \quad b = -20, \quad V_0 = \Delta S_0 + b = 5.$$

### Law of one price

The call and its replicating portfolio have the same payoff in every state, so both must cost 5 today.

## Risk-neutral probabilities encode the same restriction

Choose  $q = \mathbb{Q}(\text{up})$  so that the stock has no drift at zero interest:

$$50 = q \times 60 + (1 - q) \times 40 \implies q = \frac{1}{2}.$$

Then

$$V_0 = \mathbb{E}^{\mathbb{Q}}[X] = \frac{1}{2} \times 10 + \frac{1}{2} \times 0 = 5.$$

### Interpretation

The physical probability may differ from  $q$ . Replication determines the price first; the risk-neutral expectation is its mathematical representation.

## Student exercise: replicate a similar call

Assume a zero interest rate. A stock is worth  $S_0 = 100$  and

$$S_1 = 125 \text{ or } 75.$$

A call with strike  $K = 100$  pays 25 or 0.

1. Write the two replication equations for  $(\Delta, b)$ .
2. Find the no-arbitrage call price.
3. Find the risk-neutral up probability.
4. Explain why the physical up probability is unnecessary.

### Hint

Subtract the two payoff equations before solving for cash.

## Solution: replication and risk-neutral probability

Matching the two terminal payoffs gives

$$125\Delta + b = 25, \quad 75\Delta + b = 0.$$

Subtracting the equations,

$$\Delta = \frac{25}{125 - 75} = \frac{1}{2}, \quad b = -37.5.$$

Hence the replication cost is

$$V_0 = \Delta S_0 + b = \frac{1}{2} \times 100 - 37.5 = 12.5.$$

The risk-neutral probability solves

$$100 = 125q + 75(1 - q), \quad q = \frac{1}{2},$$

and therefore  $\mathbb{E}^{\mathbb{Q}}[X] = 12.5$ . The physical up probability is not needed.

# Replication motivates the tools used throughout the course

For a replicable payoff  $X$ , an arbitrage-free model represents its value as

$$V_t = B_t \mathbb{E}^{\mathbb{Q}} \left[ \frac{X}{B_T} \middle| \mathcal{F}_t \right].$$

To understand this expression, we now build a probability toolbox in layers:

distribution  $\longrightarrow$  expectation  $\longrightarrow$  information  $\longrightarrow$  conditional prediction  $\longrightarrow$  martingale.



## Probability toolbox

Random variables, distributions, CDFs, and densities

Expectation, variance, and dependence

Sample averages, LLN, and CLT

Characteristic functions and joint distributions

## Introductory example: a payoff is a random variable

Suppose the terminal stock price has the following physical distribution:

|              |      |      |      |
|--------------|------|------|------|
| $S_T$        | 80   | 100  | 130  |
| $\mathbb{P}$ | 0.20 | 0.50 | 0.30 |

For a call with strike  $K = 100$ ,  $X = (S_T - 100)^+$ :

|       |    |     |     |
|-------|----|-----|-----|
| $S_T$ | 80 | 100 | 130 |
| $X$   | 0  | 0   | 30  |

The payoff rule maps each state to a number; the probabilities of those numbers form the **distribution of  $X$** .

# Definitions: random variable and distribution

## Random variable and its law

A random variable is a measurable map

$$X : (\Omega, \mathcal{F}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

It assigns a numerical outcome to every state. Its distribution, or law, is

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A), \quad A \in \mathcal{B}(\mathbb{R}).$$

The random variable is the state-to-number rule; its distribution records how likely its possible values are.

# The CDF accumulates probability from left to right

## Cumulative distribution function

$$F_X(x) = \mathbb{P}(X \leq x).$$

Every CDF is nondecreasing, right-continuous, and satisfies

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow \infty} F_X(x) = 1.$$

Probabilities of intervals follow from differences:

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

### Do not confuse the objects

The **distribution** is a probability measure; the **CDF** is its cumulative function; a **density** is a derivative when one exists.

## Worked example: the exponential CDF comes from its density

Let  $X \sim \text{Exp}(\lambda)$  with

$$f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{\{x \geq 0\}}.$$

Then, for  $x \geq 0$ ,

$$F_X(x) = \int_{-\infty}^x f_X(u) du = \int_0^x \lambda e^{-\lambda u} du = 1 - e^{-\lambda x}.$$

Thus

$$\mathbb{P}(a < X \leq b) = e^{-\lambda a} - e^{-\lambda b}, \quad 0 \leq a < b.$$

When differentiability holds,

$$f_X(x) = F'_X(x).$$

## Definition: expectation is a probability-weighted average

### Expectation

For an integrable random variable  $X$ ,

$$\mathbb{E}[X] = \sum_x x \mathbb{P}(X = x) \quad \text{or} \quad \mathbb{E}[X] = \int_{\mathbb{R}} x f_X(x) dx.$$

Integrable means  $\mathbb{E}[|X|] < \infty$ , written  $X \in L^1$ .

Expectation uses a distribution to summarize the long-run or model-based average. It has the same financial unit as  $X$ .

# Functions of a random variable can be averaged directly

## Averaging a transformation

For measurable  $g$ , whenever the expectation exists,

$$\mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x) \quad \text{or} \quad \mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x)f_X(x) dx.$$

## Indicator payoff

For  $X = \mathbf{1}_A$ ,

$$\mathbb{E}[X] = \mathbb{P}(A).$$

This is why a digital payoff is naturally connected to an event probability.

## Proposition: expectation is linear and respects order

For  $X, Y \in L^1$  and  $a, b \in \mathbb{R}$ ,

### Fundamental properties

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y],$$

$$X \geq 0 \implies \mathbb{E}[X] \geq 0,$$

$$X \leq Y \implies \mathbb{E}[X] \leq \mathbb{E}[Y],$$

$$|\mathbb{E}[X]| \leq \mathbb{E}[|X|].$$

Linearity does not require independence. Independence matters when variances and products are calculated.



# Variance measures dispersion around the mean

## Mean, second moment, and variance

$$\mu = \mathbb{E}[X], \quad m_2 = \mathbb{E}[X^2], \quad \text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

The standard deviation is  $\text{SD}(X) = \sqrt{\text{Var}(X)}$ .

## Useful identity

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

Mean describes location; variance describes uncertainty around that location.

# Covariance links dependence to portfolio variance

## Covariance

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

## Variance of a linear combination

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

If  $X$  and  $Y$  are independent, then  $\text{Cov}(X, Y) = 0$ .

This identity explains why dependence matters when risks are combined.

## Reference: classical means and variances to know

| Law of $X$  | Parameters      | $\mathbb{E}[X]$ | $\text{Var}(X)$ |
|-------------|-----------------|-----------------|-----------------|
| Bernoulli   | $p$             | $p$             | $p(1 - p)$      |
| Binomial    | $n, p$          | $np$            | $np(1 - p)$     |
| Poisson     | $\lambda$       | $\lambda$       | $\lambda$       |
| Uniform     | $[a, b]$        | $(a + b)/2$     | $(b - a)^2/12$  |
| Exponential | rate $\lambda$  | $1/\lambda$     | $1/\lambda^2$   |
| Gaussian    | $\mu, \sigma^2$ | $\mu$           | $\sigma^2$      |

The exercise sheet contains the detailed derivations; class time focuses on choosing the right representation.

## Example A: derive discrete moments together

Let  $B_1, \dots, B_n$  be independent Bernoulli( $p$ ) variables and set

$$N_n = \sum_{k=1}^n B_k.$$

1. Use  $B_k^2 = B_k$  to find  $\mathbb{E}[B_k]$  and  $\text{Var}(B_k)$ .
2. Use linearity to find  $\mathbb{E}[N_n]$ .
3. Where is independence used to find  $\text{Var}(N_n)$ ?

### Progressive hint

Write the variance of the sum using covariance terms before setting them to zero.

## Solution: moments of a Bernoulli sum

For  $B_k \sim \text{Bernoulli}(p)$ ,

$$B_k^2 = B_k, \quad \mathbb{E}[B_k] = p, \quad \text{Var}(B_k) = p - p^2 = p(1 - p).$$

Linearity gives

$$\mathbb{E}[N_n] = \sum_{k=1}^n \mathbb{E}[B_k] = np.$$

Independence makes all cross-covariances vanish, so

$$\text{Var}(N_n) = \sum_{k=1}^n \text{Var}(B_k) = np(1 - p).$$

### Where independence enters

It is not needed for the mean; it is used when simplifying the variance of the sum.

## Exercise B: derive continuous moments independently

Choose one variable and justify every step.

1. If  $U \sim \text{Unif}[a, b]$ , compute  $\mathbb{E}[U]$ ,  $\mathbb{E}[U^2]$ , and  $\text{Var}(U)$ .
2. If  $T \sim \text{Exp}(\lambda)$ , compute the same three quantities using integration by parts.
3. For  $Z \sim \mathcal{N}(0, 1)$ , use symmetry to identify which odd moments vanish.

### Extension

For  $X \sim \text{Poisson}(\lambda)$ , start from  $\mathbb{E}[X(X - 1)]$  rather than expanding  $X^2$  directly.

## Solution: classical moment calculations

| Law                          | $\mathbb{E}[X]$     | $\mathbb{E}[X^2]$          | $\text{Var}(X)$       |
|------------------------------|---------------------|----------------------------|-----------------------|
| $U \sim \text{Unif}[a, b]$   | $\frac{a+b}{2}$     | $\frac{a^2 + ab + b^2}{3}$ | $\frac{(b-a)^2}{12}$  |
| $T \sim \text{Exp}(\lambda)$ | $\frac{1}{\lambda}$ | $\frac{2}{\lambda^2}$      | $\frac{1}{\lambda^2}$ |

For  $Z \sim \mathcal{N}(0, 1)$ , symmetry gives

$$\mathbb{E}[Z^{2m+1}] = 0 \quad (m \geq 0).$$

For  $X \sim \text{Poisson}(\lambda)$ ,

$$\mathbb{E}[X(X-1)] = \lambda^2 \implies \mathbb{E}[X^2] = \lambda^2 + \lambda, \quad \text{Var}(X) = \lambda.$$

## A sample summarizes repeated observations

Let  $X_1, \dots, X_n$  be observations of the same quantity. Define

### Empirical mean and empirical variance

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad v_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

$\bar{X}_n$  estimates the population mean and  $v_n$  its variance. The unbiased sample variance is

$$s_n^2 = \frac{n}{n-1} v_n.$$

These statistics will later summarize samples of returns, losses, and simulated payoffs.



# The LLN finds the target; the CLT describes the error

Let  $X_1, X_2, \dots$  be i.i.d. with mean  $\mu$ .

## Law of Large Numbers

If  $\mathbb{E}[|X_1|] < \infty$ , then

$$\bar{X}_n \longrightarrow \mu = \mathbb{E}[X_1] \quad \text{in probability.}$$

If also  $0 < \sigma^2 = \text{Var}(X_1) < \infty$ ,

## Central Limit Theorem

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow[n \rightarrow \infty]{} \mathcal{N}(0, 1) \quad \text{in law.}$$

The LLN says the error vanishes; the CLT says its typical size is  $\sigma/\sqrt{n}$  and gives its approximate shape.

## Worked example: estimate a success probability

Let  $X_i \sim \text{Bernoulli}(0.60)$  independently and let  $\bar{X}_{100}$  be the success frequency in 100 trials.

$$\mathbb{E}[\bar{X}_{100}] = 0.60, \quad \text{SD}(\bar{X}_{100}) = \sqrt{\frac{0.60(0.40)}{100}} \approx 0.049.$$

The LLN explains why the frequency should be near 0.60. The CLT gives

$$\mathbb{P}(0.50 \leq \bar{X}_{100} \leq 0.70) \approx \mathbb{P}(-2.04 \leq Z \leq 2.04) \approx 0.96.$$

### Why it matters

Averages stabilize, but their residual sampling uncertainty is of order  $1/\sqrt{n}$ .

# Characteristic functions encode an entire distribution

## Characteristic function

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \quad u \in \mathbb{R}.$$

- $|e^{iuX}| = 1$ , so  $\varphi_X$  always exists.
- The characteristic function uniquely determines the distribution.
- It turns probability calculations into Fourier calculations.
- Unlike a moment generating function, it does not require exponential moments.

This is a recognition tool for later work, not a full Fourier theory.

## Independent sums become products in Fourier space

If  $X$  and  $Y$  are independent,

### Product rule

$$\varphi_{X+Y}(u) = \varphi_X(u)\varphi_Y(u).$$

For  $Z \sim \mathcal{N}(0, 1)$ ,

$$\varphi_Z(u) = e^{-u^2/2}.$$

Thus, for independent standard normals  $Z_1, Z_2$ ,

$$\varphi_{Z_1+Z_2}(u) = e^{-u^2},$$

which identifies  $Z_1 + Z_2 \sim \mathcal{N}(0, 2)$ .

## Joint distributions describe dependence between variables

Consider

|                | $Y = 0$ | $Y = 1$ | $\mathbb{P}_X$ |
|----------------|---------|---------|----------------|
| $X = 0$        | 0.30    | 0.10    | 0.40           |
| $X = 1$        | 0.20    | 0.40    | 0.60           |
| $\mathbb{P}_Y$ | 0.50    | 0.50    | 1              |

- The four interior values form the **joint distribution**.
- Row and column sums are the **marginal distributions**.
- The joint law tells us how observing one variable changes what we know about the other.

This observation creates the need for information and conditional distributions.

## Information and conditional expectation

Information first, formalism second

Conditional distributions and conditional expectation

Gaussian conditioning

Properties used in dynamic pricing

## Introductory example: observing a signal reveals a partition

A fair die produces the full outcome  $X(\omega) = \omega$ . Instead of observing  $X$ , we observe only

$$Y = \mathbf{1}_{\{X \text{ is even}\}}.$$

Knowing  $Y$  tells us which information set contains the outcome:

$$O = \{1, 3, 5\}, \quad E = \{2, 4, 6\}.$$

- $Y$  distinguishes odd from even outcomes.
- $Y$  cannot distinguish 2 from 4 or 6.
- Observing  $Y$  therefore reveals the partition  $\{O, E\}$ .

## The notation $\sigma(Y)$ records exactly what $Y$ reveals

After observing the parity signal  $Y$ , we can decide whether

$$\omega \in O, \quad \omega \in E, \quad \omega \in \Omega, \quad \omega \in \emptyset.$$

We cannot decide whether  $\omega = 2$ , because  $Y$  gives the same signal for 2, 4, and 6.

### Information generated by a random variable

$$\sigma(Y) = \{\text{events whose occurrence can be determined from } Y\}.$$

For this example,

$$\sigma(Y) = \{\emptyset, O, E, \Omega\}.$$

Thus  $\sigma(Y)$  is the information revealed by observing  $Y$ , written as a collection of observable events.



# A sigma-algebra formalizes observable yes-or-no questions

## $\sigma$ -algebra

A collection  $\mathcal{G}$  of events is a  $\sigma$ -algebra if

- $\Omega \in \mathcal{G}$ ;
- $A \in \mathcal{G}$  implies  $A^c \in \mathcal{G}$ ;
- $A_1, A_2, \dots \in \mathcal{G}$  implies  $\bigcup_n A_n \in \mathcal{G}$ .

The closure rules ensure that observable questions can be combined consistently.

# Measurable means determined by the available information

## $\mathcal{G}$ -measurable random variable

A random variable  $Z$  is  $\mathcal{G}$ -measurable if

$$\{Z \in B\} \in \mathcal{G} \quad \text{for every Borel set } B.$$

## Information interpretation

“ $Z$  is  $\mathcal{G}$ -measurable” means that the information in  $\mathcal{G}$  is sufficient to determine  $Z$ .

In a finite model,

$$Z \text{ is } \sigma(Y)\text{-measurable} \iff Z = g(Y) \text{ for some function } g.$$

Thus  $Z$  must take the same value whenever  $Y$  takes the same value.

## Worked example: measurable quantities are constant on signal sets

For the parity signal, a  $\sigma(Y)$ -measurable variable has the form

$$Z = a\mathbf{1}_O + b\mathbf{1}_E.$$

- $Z_1 = 10 \cdot \mathbf{1}_E$  is measurable: parity determines it.
- $Z_2 = X$  is not measurable: parity does not determine the die value.
- $Z_3 = (-1)^X$  is measurable: it is a function of parity.

### Finite-state test

A variable is measurable with respect to the observed partition if and only if it is constant on every information set.

## Student exercise: what does a coarse price signal reveal?

A terminal stock price takes values 80, 100, or 130. The signal is

$$Y = \mathbf{1}_{\{S_T \geq 100\}}.$$

1. Write the partition generated by observing  $Y$ .
2. List the events in  $\sigma(Y)$ .
3. Is the call payoff  $(S_T - 100)^+$  measurable with respect to  $\sigma(Y)$ ?
4. Is the digital payoff  $\mathbf{1}_{\{S_T \geq 100\}}$  measurable?

### Test

Can two states with the same signal produce different values of the quantity?

## Solution: the signal groups states with the same observation

The signal distinguishes the two information sets

$$\{80\} \quad \text{and} \quad \{100, 130\}.$$

Hence

$$\sigma(Y) = \{\emptyset, \Omega, \{80\}, \{100, 130\}\}.$$

The call payoff takes values 0 and 30 inside  $\{100, 130\}$ , so it is **not**  $\sigma(Y)$ -measurable.

The digital payoff equals the signal:

$$\mathbf{1}_{\{S_T \geq 100\}} = Y,$$

so it is  $\sigma(Y)$ -measurable.

### Test

The signal determines a quantity exactly if that quantity is constant on every information set.

## A conditional distribution updates the relevant probabilities

Return to the joint distribution

|         | $Y = 0$ | $Y = 1$ |
|---------|---------|---------|
| $X = 0$ | 0.30    | 0.10    |
| $X = 1$ | 0.20    | 0.40    |

Since  $\mathbb{P}(Y = 1) = 0.50$ ,

$$\mathbb{P}(X = 1 \mid Y = 1) = \frac{\mathbb{P}(X = 1, Y = 1)}{\mathbb{P}(Y = 1)} = \frac{0.40}{0.50} = 0.80.$$

Observing  $Y = 1$  replaces the marginal law of  $X$  by the conditional law of  $X$  given that information.

# Conditional expectation is the best prediction from current information

## Information interpretation

$\mathbb{E}[X \mid \mathcal{G}]$  is the prediction of  $X$  based only on the information in  $\mathcal{G}$ .

It must be  $\mathcal{G}$ -measurable because a prediction cannot use information that has not been observed.

Conditioning on a random variable is shorthand:

$$\mathbb{E}[X \mid Y] := \mathbb{E}[X \mid \sigma(Y)].$$

In a finite partition, the prediction is the average of  $X$  inside the observed information set.

## Definition: conditional expectation matches averages on known events

### Conditional expectation

For  $X \in L^1$  and  $\mathcal{G} \subseteq \mathcal{F}$ ,  $\mathbb{E}[X \mid \mathcal{G}]$  is the almost surely unique  $\mathcal{G}$ -measurable random variable  $Z$  such that

$$\int_A Z \, d\mathbb{P} = \int_A X \, d\mathbb{P} \quad \text{for every } A \in \mathcal{G}.$$

### Finite partition rule

On each information set  $A$  with positive probability,

$$\mathbb{E}[X \mid \mathcal{G}](\omega) = \mathbb{E}[X \mid A], \quad \omega \in A.$$



## Example A: compute a prediction from a partial signal

On  $\Omega = \{1, 2, 3, 4\}$  with equal probabilities, let

$$X(1) = 0, \quad X(2) = 4, \quad X(3) = 8, \quad X(4) = 12,$$

and let  $Y = L$  on  $\{1, 2\}$  and  $Y = H$  on  $\{3, 4\}$ .

1. What partition and sigma-algebra does  $Y$  reveal?
2. Which values can  $\mathbb{E}[X \mid Y]$  take?
3. Average  $X$  inside each information set.

Working together gives

$$\mathbb{E}[X \mid Y] = 2 \cdot \mathbf{1}_{\{Y=L\}} + 10 \cdot \mathbf{1}_{\{Y=H\}}.$$

Same information implies the same conditional prediction.

## Exercise B: value a payoff after a coarse signal

A fair die is rolled and

$$X = 12 \cdot \mathbf{1}_{\{\omega=6\}}.$$

You observe only whether

$$L = \{1, 2, 3, 4\} \quad \text{or} \quad H = \{5, 6\}$$

occurred.

1. Write  $\sigma(L, H)$ .
2. Compute  $\mathbb{E}[X \mid L]$  and  $\mathbb{E}[X \mid H]$ .
3. Write the random variable  $\mathbb{E}[X \mid \sigma(L, H)]$ .
4. Check that its expectation equals  $\mathbb{E}[X]$ .

## Solution: average the payoff inside each information set

The revealed sigma-algebra is

$$\sigma(L, H) = \{\emptyset, L, H, \Omega\}.$$

On  $L$ , the payoff is always zero. On  $H = \{5, 6\}$ , its two values are 0 and 12. Therefore

$$\mathbb{E}[X \mid L] = 0, \quad \mathbb{E}[X \mid H] = 6,$$

and

$$\mathbb{E}[X \mid \sigma(L, H)] = 6 \cdot \mathbf{1}_H.$$

Finally,

$$\mathbb{E}[\mathbb{E}[X \mid \sigma(L, H)]] = 6 \times \frac{2}{6} = 2 = \mathbb{E}[X].$$

## A joint density turns conditioning into a calculation

Suppose  $(X, Y)$  has joint density  $p_{X,Y}$ . For  $p_Y(y) > 0$ ,

$$\begin{aligned}p_Y(y) &= \int_{\mathbb{R}} p_{X,Y}(x, y) dx, \\p_{X|Y}(x | y) &= \frac{p_{X,Y}(x, y)}{p_Y(y)}, \\g(y) &= \int_{\mathbb{R}} f(x) p_{X|Y}(x | y) dx = \mathbb{E}[f(X) | Y = y].\end{aligned}$$

Consequently,

$$\boxed{\mathbb{E}[f(X) | Y] = g(Y)}.$$

### Measurability check

$g(Y)$  is a function of the observed signal  $Y$ , so it is  $\sigma(Y)$ -measurable.

## Worked example: conditioning with a joint density

Let

$$f_{X,Y}(x,y) = 2 \cdot \mathbf{1}_{\{0 < x < y < 1\}}.$$

For  $0 < y < 1$ , first compute the marginal density:

$$f_Y(y) = \int_0^y 2 \, dx = 2y.$$

Then

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{1}{y} \mathbf{1}_{(0,y)}(x).$$

Therefore

$$\mathbb{E}[X | Y = y] = \int_0^y x \frac{1}{y} \, dx = \frac{y}{2}.$$

Thus

$$\mathbb{E}[X | Y] = \frac{Y}{2},$$

which is a function of  $Y$ .

## Student exercise: reverse the triangular density

Let

$$f_{X,Y}(x,y) = 2 \cdot \mathbf{1}_{\{0 < y < x < 1\}}.$$

For a fixed  $0 < y < 1$ :

1. Draw or describe the support of  $(X, Y)$ .
2. Compute  $f_Y(y)$  by integrating out  $x$ .
3. Find  $f_{X|Y}(x | y)$  and its support.
4. Compute  $\mathbb{E}[X | Y = y]$ .

### Hint

For fixed  $y$ , the admissible values of  $x$  run from  $y$  to 1.

## Solution: conditioning on the reversed triangle

For fixed  $0 < y < 1$ ,

$$p_Y(y) = \int_y^1 2 \, dx = 2(1 - y).$$

Hence

$$p_{X|Y}(x | y) = \frac{2}{2(1 - y)} \mathbf{1}_{(y,1)}(x) = \frac{1}{1 - y} \mathbf{1}_{(y,1)}(x).$$

Therefore

$$\mathbb{E}[X | Y = y] = \int_y^1 \frac{x}{1 - y} \, dx = \frac{1 + y}{2},$$

and, as a random variable,

$$\mathbb{E}[X | Y] = \frac{1 + Y}{2}.$$

## Gaussian conditioning remains Gaussian

Suppose  $(X, Y)$  is jointly Gaussian, with means  $\mu_X, \mu_Y$ , standard deviations  $\sigma_X, \sigma_Y > 0$ , and correlation  $\rho$ . Then

$$X \mid (Y = y) \sim \mathcal{N}\left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y), \sigma_X^2(1 - \rho^2)\right).$$

Consequently,

$$\mathbb{E}[X \mid Y] = \mu_X + \rho \frac{\sigma_X}{\sigma_Y}(Y - \mu_Y), \quad \text{Var}(X \mid Y) = \sigma_X^2(1 - \rho^2).$$

### Why?

With  $a = \text{Cov}(X, Y) / \text{Var}(Y)$ , the residual  $R = X - \mu_X - a(Y - \mu_Y)$  is uncorrelated with  $Y$ . Joint Gaussianity makes  $R$  independent of  $Y$ , which gives both formulas.

Larger  $|\rho|$  means that observing  $Y$  removes more uncertainty about  $X$ .



# Conditional expectation respects algebra and information

For integrable  $X, Y$  and constants  $a, b$ :

## Linearity, order, and information

$$\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}],$$

$$X \leq Y \implies \mathbb{E}[X \mid \mathcal{G}] \leq \mathbb{E}[Y \mid \mathcal{G}].$$

If  $X$  is  $\mathcal{G}$ -measurable, then  $\mathbb{E}[X \mid \mathcal{G}] = X$ . If  $X$  is independent of  $\mathcal{G}$ , then  $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$ .

## Known factors come out and predictions satisfy the tower rule

Let  $Z$  be  $\mathcal{G}$ -measurable with  $ZX \in L^1$ , and let  $\mathcal{H} \subseteq \mathcal{G}$ .

### Taking out what is known

$$\mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}].$$

### Tower property

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}], \quad \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X].$$

Re-predicting with less information is equivalent to predicting directly from that less detailed information.

## Processes and martingales

Information evolving through time

Random walks and martingales

# A stochastic process is a collection of dated random variables

## Discrete-time stochastic process

A process  $(X_n)_{n \geq 0}$  is a sequence of random variables defined on the same probability space.

There are two complementary viewpoints:

- fix  $n$ :  $X_n$  is one random variable with a distribution;
- fix  $\omega$ :  $n \mapsto X_n(\omega)$  is one sample path.

The same tools now have a time index: distributions describe each date, joint laws describe dependence across dates, and information determines what can be predicted.

## Coin tosses show how information grows over time

Let  $\xi_1, \xi_2$  be successive coin-toss outcomes and let  $S_n$  count heads by time  $n$ .

| Time | Observed variables | Information available          |
|------|--------------------|--------------------------------|
| 0    | none               | no toss is known               |
| 1    | $\xi_1$            | first toss is known            |
| 2    | $\xi_1, \xi_2$     | full two-toss history is known |

The information sets become finer as observations arrive. A prediction at time 1 may use  $\xi_1$ , but not  $\xi_2$ .

# Filtrations record information and adapted processes respect it

## Filtration

A filtration  $(\mathcal{F}_n)$  is an increasing family of sigma-algebras:

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots .$$

Here  $\mathcal{F}_n$  is the information available up to time  $n$ .

## Adapted process

$(X_n)$  is adapted if  $X_n$  is  $\mathcal{F}_n$ -measurable for every  $n$ .

Adapted means no look-ahead: today's value or decision can use the observed past and present, never the future.

## Student exercise: detect look-ahead

Two fair coin tosses are encoded by  $\xi_1, \xi_2 \in \{0, 1\}$  and

$$\mathcal{F}_0 = \{\emptyset, \Omega\}, \quad \mathcal{F}_1 = \sigma(\xi_1), \quad \mathcal{F}_2 = \sigma(\xi_1, \xi_2).$$

Consider

$$X_0 = 0, \quad X_1 = \xi_1 + \xi_2, \quad X_2 = \xi_1 + \xi_2.$$

1. Is  $(X_0, X_1, X_2)$  adapted to  $(\mathcal{F}_n)$ ?
2. Replace  $X_1$  by a natural adapted value while keeping  $X_2$  unchanged.

## Solution: time 1 cannot use the second toss

$X_0$  is constant and hence  $\mathcal{F}_0$ -measurable. The terminal value  $X_2 = \xi_1 + \xi_2$  is  $\mathcal{F}_2$ -measurable.

At time 1, however,  $X_1$  depends on  $\xi_2$ , which is not yet observed. Thus  $X_1$  is not  $\mathcal{F}_1$ -measurable and the process is not adapted.

The cumulative number of heads

$$\tilde{X}_0 = 0, \quad \tilde{X}_1 = \xi_1, \quad \tilde{X}_2 = \xi_1 + \xi_2$$

is adapted.

### Diagnostic

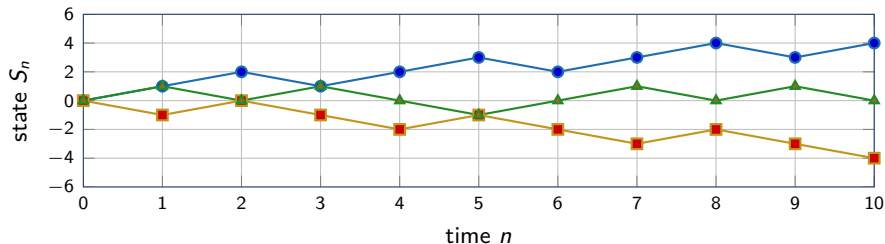
At each date, check whether the value is determined by the information available at that date.



# A random walk evolves through many possible sample paths

Let  $\varepsilon_k \in \{-1, +1\}$  and

$$S_n = S_0 + \sum_{k=1}^n \varepsilon_k.$$



One model generates many possible paths; only one is observed in a realized history.

# Independence makes random-walk moments tractable

Let the increments be i.i.d. with

$$\mathbb{E}[\varepsilon_k] = m, \quad \text{Var}(\varepsilon_k) = \sigma^2.$$

## Random-walk moments

$$\mathbb{E}[S_n] = S_0 + nm, \quad \text{Var}(S_n) = n\sigma^2.$$

For a symmetric  $\pm 1$  walk,  $m = 0$  and  $\sigma^2 = 1$ . The mean stays at  $S_0$  while uncertainty grows linearly with time.

# A martingale is a fair conditional prediction

## Information interpretation

Given everything known today, the best prediction of tomorrow's value is today's value.

## Martingale

An integrable adapted process  $(M_n)$  is a martingale if

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n.$$

This is the mathematical form of a fair game: past wins and losses are known, but no predictable gain remains.

## Worked proof: the centered random walk is a martingale

Let

$$M_n = \sum_{k=1}^n \varepsilon_k,$$

where the increments are independent, centered, and  $\mathcal{F}_n = \sigma(\varepsilon_1, \dots, \varepsilon_n)$ .

Then  $M_n$  is known at time  $n$ , while  $\varepsilon_{n+1}$  is independent of  $\mathcal{F}_n$ :

$$\begin{aligned}\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[M_n + \varepsilon_{n+1} \mid \mathcal{F}_n] \\ &= M_n + \mathbb{E}[\varepsilon_{n+1}] \\ &= M_n.\end{aligned}$$

This calculation uses measurability, independence, conditional expectation, and filtration in one place.

## Student exercise: find a second random-walk martingale

Let  $S_n = \sum_{k=1}^n \varepsilon_k$ , where the independent increments satisfy

$$\mathbb{P}(\varepsilon_k = 1) = \mathbb{P}(\varepsilon_k = -1) = \frac{1}{2}, \quad \mathcal{F}_n = \sigma(\varepsilon_1, \dots, \varepsilon_n).$$

Show that

$$M_n = S_n^2 - n$$

is a martingale.

### Hint

Expand  $S_{n+1}^2 = (S_n + \varepsilon_{n+1})^2$  and condition on  $\mathcal{F}_n$ .

## Solution: subtract the predictable variance growth

Since  $S_n$  is  $\mathcal{F}_n$ -measurable and  $\varepsilon_{n+1}$  is independent of  $\mathcal{F}_n$ ,

$$\begin{aligned}\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[(S_n + \varepsilon_{n+1})^2 - (n+1) \mid \mathcal{F}_n] \\ &= S_n^2 + 2S_n\mathbb{E}[\varepsilon_{n+1}] + \mathbb{E}[\varepsilon_{n+1}^2] - (n+1) \\ &= S_n^2 - n = M_n.\end{aligned}$$

### Interpretation

$S_n^2$  grows predictably by one unit per step; subtracting  $n$  removes that drift.

# Martingales connect fair games to financial pricing

- In a fair casino game, the conditional expected future wealth equals current wealth.
- Under a risk-neutral measure  $\mathbb{Q}$ , discounted traded asset prices are martingales.
- Therefore a replicable payoff satisfies

$$\frac{V_t}{B_t} = \mathbb{E}^{\mathbb{Q}} \left[ \frac{X}{B_T} \mid \mathcal{F}_t \right].$$

Replication supplies the economics; the martingale property supplies the dynamic conditional-expectation formula.

# Short differential-calculus review

Short differential-calculus review

Derivatives needed later in the course



# The derivative is defined by a limiting difference quotient

## Derivative of $g : \mathbb{R} \rightarrow \mathbb{R}$

If the limit exists, the derivative of  $g$  at  $x$  is

$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}.$$

It measures the first-order change:

$$g(x+h) = g(x) + g'(x)h + o(h).$$

For example, applying the definition to  $g(x) = x^2$  gives  $g'(x) = 2x$ .

# Partial derivatives vary one coordinate at a time

Let  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  and let  $e_i$  be the  $i$ th coordinate vector.

## Partial derivative and gradient

$$\partial_i f(x) = \lim_{h \rightarrow 0} \frac{f(x + he_i) - f(x)}{h}, \quad \nabla f(x) = (\partial_1 f(x), \dots, \partial_d f(x)).$$

If the second partial derivatives exist, the Hessian is

## Hessian

$$\nabla^2 f(x) = \left( \frac{\partial^2 f}{\partial x_i \partial x_j}(x) \right)_{1 \leq i, j \leq d}.$$

The gradient gives first-order sensitivities; the Hessian records curvature and interactions.

## Worked example: compute the gradient and Hessian

For

$$f(x, y) = x^2y + e^y,$$

the first partial derivatives are

$$f_x = 2xy, \quad f_y = x^2 + e^y, \quad \nabla f(x, y) = (2xy, x^2 + e^y).$$

Differentiating once more gives

$$f_{xx} = 2y, \quad f_{xy} = f_{yx} = 2x, \quad f_{yy} = e^y,$$

and therefore

$$\nabla^2 f(x, y) = \begin{pmatrix} 2y & 2x \\ 2x & e^y \end{pmatrix}.$$

# The chain rule connects sensitivities to value changes

If  $x(t) \in \mathbb{R}^d$  is differentiable and  $f$  is differentiable, then

## Multivariable chain rule

$$\frac{d}{dt}f(x(t)) = \nabla f(x(t)) \cdot x'(t).$$

For an option value  $V = V(t, S)$  and a differentiable price path  $S = S(t)$ ,

$$\frac{d}{dt}V(t, S(t)) = V_t(t, S(t)) + V_S(t, S(t))S'(t).$$

Here  $V_S$  measures sensitivity to the underlying price,  $V_t$  measures sensitivity to calendar time, and  $V_{SS}$  measures curvature in the underlying price.